Predicting India's Weighted Average Call Money Rate

A Regime-Aware Analysis of India's Overnight Call Money Rate

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545
weekly observations
Feb 2014 – Jul 2024

117
features
across 10 sources

$\sim$10 bp
forecast RMSE
389 OOS predictions

70.9%
directional accuracy
up vs. down moves

2
structural regimes
discovered, not assumed

The Problem: Forecasting WACMR is Genuinely Hard

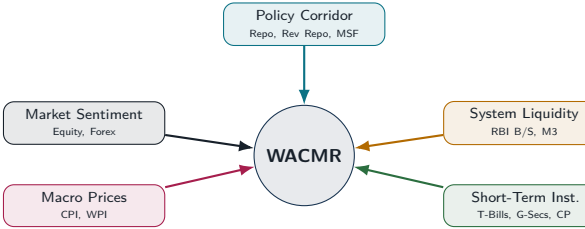
What is WACMR?

- Volume-weighted overnight interbank lending rate.
- The **operational target** of the RBI's Liquidity Adjustment Facility.
- Moves *first* when liquidity tightens or loosens — before the rest of the rate curve follows.

Who relies on it?

- **RBI**: policy enforcement & liquidity monitoring.
- **Treasuries & funds**: overnight pricing, NAV decisions.
- **Researchers**: regime analysis & monetary transmission.

Five competing transmission channels



Core challenge: the relative weight of these five channels shifts across structural regimes — a non-stationary, multi-channel forecasting problem.

Four Hypotheses We Test

H1 — Regime structure exists

India's money market data contains ≥ 2 statistically distinct macroeconomic regimes that K-Means can recover *without* being given any date labels.

H2 — Regime labels improve forecasts

Injecting regime indicators into a supervised model reduces out-of-sample RMSE by $\geq 15\%$ versus an identical baseline.

H3 — Corridor rates dominate

Repo, Reverse Repo, and MSF emerge as the top SHAP predictors — WACMR is, mechanically, a corridor-bound rate.

H4 — Markets add value

Adding 28 Nifty / USD-INR features materially improves forecast accuracy and contributes meaningful SHAP signal.

Each hypothesis maps to a falsifiable test. Verdicts return on slide 12.

Data Foundation: 10 Sources, 545 Friday-Aligned Weeks

NDAP API — 8 RBI datasets

Domain	Dataset
Monetary Policy	Ratios & Rates
Central Bank B/S	Liab. & Assets
Money Supply	Weekly Aggregates
Debt Markets	CP / T-Bills / G-Secs
Repo Markets	Market Repo Trans.
Inflation	Major Price Indices

Yahoo Finance — 2 datasets

- Nifty 50 weekly OHLCV
- USD/INR weekly OHLCV

Curated NLP corpus

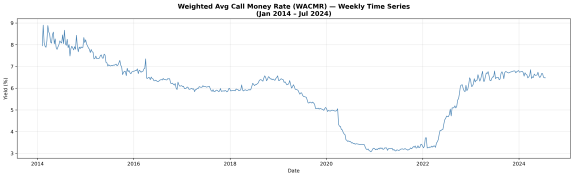
- 75 RBI policy events
- Feb 2014 – Aug 2024
- 6 categories, sentiment $\in [-1, +1]$

The Friday-alignment problem

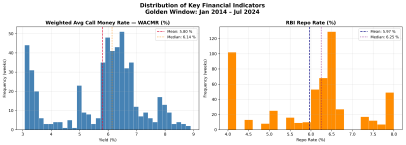
- Three different date formats across NDAP (ISO, FY week, CY week).
- Custom parsers snap every series to the **nearest following Friday**.
- Without this, dates drift up to **12 weeks** for FY tables.

Output: 545 rows $\times$ 119 columns, SQLite (577 KB), zero missing on target.

EDA: A Bimodal Story Visible to the Naked Eye



Taper-tantrum peak (8.90%), demonetisation, COVID, 2022 re-tightening.



Density is clearly bimodal — two operating modes.

WACMR descriptive statistics

5.80% Mean	**1.46%** Std Dev
3.07% Min (Dec 2020)	**8.90%** Max (Feb 2014)

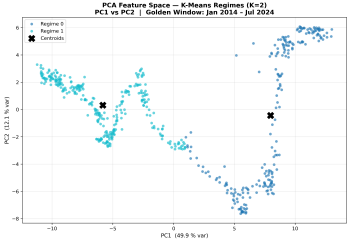
Top non-corridor correlations

Feature	Domain	Pearson r
CBLO Rate	Debt	**0.96**
CD Rate	Debt	**0.85**
CP Outstanding	CP	0.42
USD/INR	Forex	−0.18
Nifty 50	Equity	−0.27

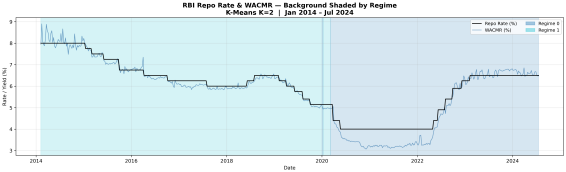
Early signal: equity and forex are weak in raw correlations.

Finding 1 — Two Structural Regimes Discovered

H1 ✓



Two clusters cleanly separated along PC1 (49.9% variance).



Method: 14-PC PCA (90.4% var) → K-Means sweep K=2–7.

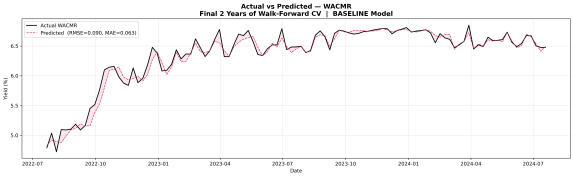
K=2 wins on silhouette (0.4238) *and* interpretability.

	R1 — Tightening	**R0 — Accommodation**
Weeks	315	230
Period	Feb '14 → Feb '20	Jan '20 → Jul '24
Avg WACMR	**6.53%**	**4.81%**
Avg Repo	6.59%	5.12%
Avg MSF	7.15%	5.37%
Contains	taper-tantrum, demon., NBFC	COVID cuts, SDF, 2022–23 hikes

Insight: even the 2022–23 hiking cycle stays in R0 — the operational framework changed *permanently* after 2020. The 1.72-pp gap is structural.

Finding 2 — ∼10 bp Forecasts From Public Data

H2 ×



389 walk-forward predictions. Misses concentrated at COVID cut and 2022 hikes.

0.1019
RMSE
∼10 basis points

0.0646
MAE

70.9%
Directional Acc.
up vs down

Walk-forward XGBoost setup

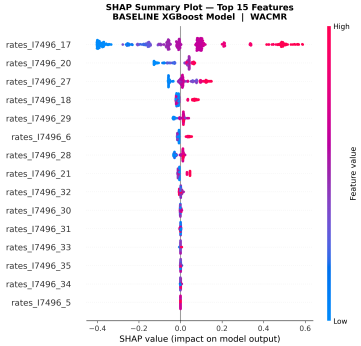
- **Train:** expanding window starting at week 156.
- **Test:** one-week-ahead, 389 OOS predictions.
- **Baseline:** 117 features. **Regime-aware:** +5 regime features.

Metric	Baseline	Regime-Aware	Δ
RMSE	0.1019	0.1044	+2.5% worse
MAE	0.0646	0.0646	0.0%
Dir. Acc.	70.9%	70.9%	+0.0pp

H2 rejected: explicit regime tagging adds nothing.
Why: XGBoost already learns the regime structure implicitly through corridor rates and AR lags — a one-hot label is redundant.

Finding 3 — Corridor Dominates; Markets Don't

H3 ✓ H4 ×



Permutation SHAP, top 15 features.

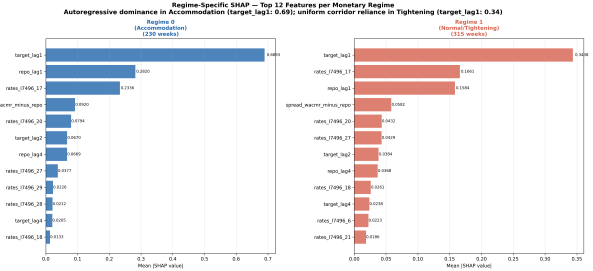
What SHAP tells us

- **Top 3 features** contribute SHAP = 0.895 of the 1.285 total in top 15 (∼70%).
- Top 15 = corridor rates, AR lags, one engineered spread, one balance-sheet item.
- **Equity / forex: zero appearances** in top 15.
- First market feature: rank 23, SHAP = 0.0038 (∼100× smaller than corridor).

H3 confirmed: WACMR is mechanically corridor-bound. Repo, Reverse Repo and MSF do the heavy lifting.

H4 rejected: Nifty and USD/INR carry no incremental information for overnight rate forecasting once corridor and lags are in the model.

Finding 4 — Drivers Shift Across Regimes



Accommodation regime (230 wks)

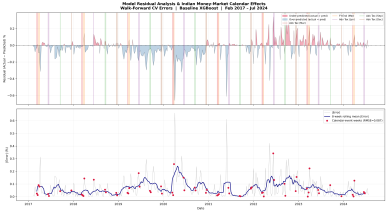
- `target_lag1` dominant SHAP = **0.689**.
- Repo (current + lag) are the next-strongest drivers.
- Persistence is the binding force.

Tightening regime (315 wks)

- `target_lag1` drops to 0.344 (∼50% lower).
- All driver SHAPs ∼50% smaller.
- RBI intervention disrupts the autoregression.

Insight: all SHAP magnitudes are $\sim 2\times$ larger in Accommodation — WACMR is meaningfully *more predictable* in surplus than during active tightening.

Finding 5 — Errors Are Policy Surprises, Not Calendar Pressure



Calendar events — NOT the problem

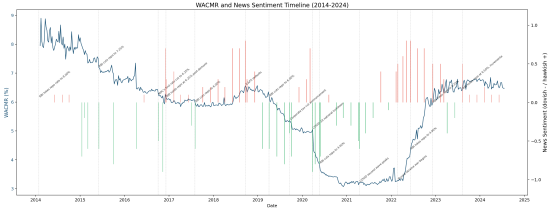
- 66 calendar-event weeks tested.
- RMSE on calendar weeks: **0.0870** (*better* than baseline).
- RMSE on non-calendar weeks: 0.1046.
- *10 years of training learns seasonal patterns.*

Policy surprises — the real problem

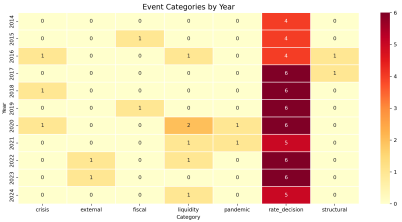
- COVID inter-meeting cut, Apr 2020: −0.66% miss.
- Surprise hike, May 2022: −0.42% miss.
- Post-hike reverberations, 2022: ±0.30–0.34%.
- *Unforecastable from historical features alone.*

Insight: the residual ceiling is set by RBI discretion — only forward-looking signals (event dummies, sentiment) can lift it.

Finding 6 — NLP Sentiment Validates the Story (Weakly)



75 hand-curated RBI policy events, sentiment $\in [-1, +1]$.



Event density heatmap by year and category.

Pearson r = −0.21 between weekly sentiment and WACMR change. Dovish weeks weakly co-move with falling WACMR. Too small as a primary feature, but it *validates* the qualitative event overlay around the model.

Why hand-curated?

VADER / FinBERT misclassify Indian monetary-policy language — "accommodative withdrawal" reads as *positive*.

Hypothesis Verdicts — Final Scorecard

Hypothesis	Prediction	Verdict	Key Evidence
H1 — Regime structure exists	$\geq$ 2 distinct regimes	✓**CONFIRMED**	K-Means K=2, silhouette 0.4238; aligns with COVID-era operational shift.
H2 — Regime labels improve forecasts	$\geq$ 15% RMSE reduction	× **REJECTED**	Regime-aware RMSE 0.1044 vs baseline 0.1019 (+2.5% *worse*); regime info already implicit in corridor + AR lags.
H3 — Corridor rates dominate	Top SHAP features	✓**CONFIRMED**	Repo, Reverse Repo, MSF + AR lags fill the entire top 15.
H4 — Equity/forex add value	Meaningful RMSE improvement	× **REJECTED**	First market feature at rank 23, SHAP $\sim$100$\times$ smaller than corridor; raw correlations already weak.

Two-of-four confirmed — and the rejections are the most actionable findings. A model can be built without regime labels; analyst effort on equity/forex for overnight rate work is misallocated.

Five Economic Insights From the Findings

1. The 2020 break is structural, not transitional

Even the 2022–23 hiking cycle operates inside the post-COVID framework. The shift is permanent, not a temporary COVID artefact.

2. The policy corridor explains almost everything; equity and forex explain almost nothing

Empirically confirmed via SHAP, not assumed. Stops a long-running practitioner debate.

3. Forecasting to $\sim$10 bp is feasible from open public data

No proprietary data, no intraday feeds, no LAF utilisation series — yet 70.9% directional accuracy.

4. Tail errors come from policy discontinuities, not market or calendar pressure

The residual ceiling is set by RBI discretion. Better historical features can't lift it.

5. The WACMR–Repo spread is a real-time liquidity barometer

Higher SHAP than MSF Rate itself — a clean, free signal of liquidity stress.

Policy Recommendations — Operational Levers

A. RBI — Adopt the WACMR–Repo spread as a published liquidity indicator

- **Soft alert:** spread $> +0.10\%$ for 3 consecutive weeks.
- **Hard alert:** spread $> +0.25\%$ in any week.
- *Evidence:* 4th-highest SHAP feature; bigger signal than MSF rate alone.
- **Why now:** a transparent threshold rule reduces guesswork in market commentary about RBI's reaction function.

B. Treasury / money-market desks — Size positions to regime confidence

- Use cluster-distance ratio as a regime-confidence score.
- Near the regime boundary $\rightarrow$ reduce position size.
- *Evidence:* SHAP magnitudes $\sim 2\times$ larger in Accommodation — directional bets are higher-quality there.

C. Equity / FX desks — Stop using Nifty and USD/INR for overnight rate forecasting

- Reallocate analyst time off this dead-end.
- *Evidence:* first market feature at rank 23; SHAP $\sim 100\times$ smaller than corridor rates; raw correlation $|r| \leq 0.27$.
- This is a **negative result with positive value** — it kills a costly intuition.

D. Modelling teams — Don't waste effort on explicit regime labels

- A well-tuned XGBoost on corridor + AR lags already absorbs the regime structure.
- *Evidence:* regime-aware variant performs $+2.5\%$ *worse* on RMSE.
- Spend that engineering on event dummies and forward-looking features instead.

Policy Recommendations — Data Infrastructure & Communication

E. NDAP — Publish high-value series the RBI itself references

- Weekly machine-readable T-Bill cut-off yields.
- Intraday LAF utilisation data.
- Standing Deposit Facility utilisation series.
- *Evidence:* our target substitution (we forecast WACMR rather than 91-day yields) was forced solely by NDAP gaps. These series exist in RBI publications but are not API-accessible.
- **Cost:** negligible. **Benefit:** unlocks an entire research stream on monetary transmission.

F. Policy communication — Dashboards on free infrastructure are sufficient

- Sub-second load times achieved on free-tier Vercel + Render.
- No vendor lock-in, no subscription cost.
- *Evidence:* our dashboard ships 10 pages, regime-aware forecasts, SHAP attributions and a counterfactual simulator on free infrastructure.
- **Implication:** the binding constraint on transparent monetary-policy analytics is open *data*, not compute.

G. Future research — Build forward-looking event signals

- Replace 75-event manual corpus with a FinBERT pipeline over the full RBI press-release archive (∼5,000 documents).
- Add a binary MPC-meeting dummy — the only interpretable lever against the residual ceiling.

Limitations and Future Work

Limitation	Future work
Target substitution: T-Bill yields unavailable on NDAP → switched primary target to WACMR.	Retrain the pipeline on T-Bill cut-off yields once NDAP publishes them.
Two-regime simplicity (K=3 only 0.019 worse on silhouette).	Try HDBSCAN / hierarchical clustering for finer regime structure.
17.4% imputation on 2 columns (CP outstanding, FX swap).	Sensitivity analysis without imputed columns.
Single one-week-ahead horizon only.	Multi-horizon forecasts with conformal prediction intervals.
Single-target regression.	Vector autoregression for joint dynamics across the rate curve.
No causal identification.	Synthetic control for specific policy events (e.g. SDF launch).
75-event NLP corpus is too small for a strong signal.	Scale to a FinBERT pipeline over the full RBI press-release archive.
No event dummies for surprise inter-meeting actions.	Add an MPC calendar binary dummy — the strongest available lever against the residual ceiling.

Each limitation is concrete and actionable — not a generic disclaimer. The pipeline is engineered to absorb each fix without re-architecting.

Conclusion — The One-Line Takeaway

0.1019
RMSE (∼10 bp)
baseline beats regime-aware

70.9%
Directional Accuracy
389 OOS predictions

0.4238
Silhouette Score
at K=2, regimes discovered

The 2020 structural break is real
K-Means recovered it without being told a single date.

The corridor explains WACMR; equity and forex do not
Empirically confirmed — not assumed.

Calendar pressure is not the forecasting problem
Genuine policy surprises are. The residual ceiling is set by RBI discretion.

Headline: India's overnight call-money rate is corridor-bound, regime-structured, and predictable to ∼10 bp from open public data alone — but the last basis points belong to RBI discretion, not to better features.

Thank You

Questions Welcome

Dashboard
dsm-project-phi.vercel.app

API
wacmr-api.onrender.com

GitHub
github.com/AJ112103/DSM_project

Aamer Jalan & Arnav Goyal
Ashoka University | Data Science and Management

All figures, models and policy claims are reproducible from the committed pipeline (stages 1–6).

Appendix A — Walk-Forward XGBoost Configuration

Walk-forward setup

- Initial train window: weeks 1–156 (3 years).
- Step: 1 week, expanding window.
- Test horizon: 1 week ahead.
- Total OOS predictions: **389**.
- No look-ahead; features computed from past data only.

XGBoost hyperparameters

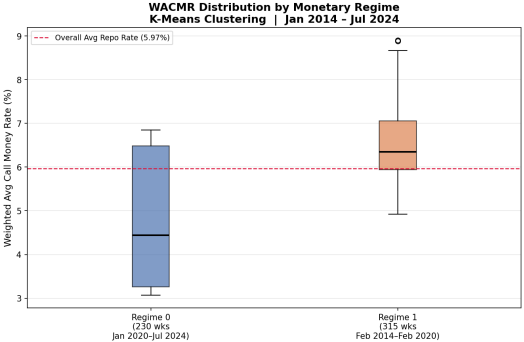
- n_estimators: 400
- learning_rate: 0.05
- max_depth: 4
- subsample: 0.8
- colsample_bytree: 0.8
- reg_alpha: 0.1, reg_lambda: 1.0
- Random seed fixed; deterministic.

Baseline: 117 features. **Regime-aware:** 122 features (+ 2 one-hot regime indicators, + 2 cluster-distance features, + 1 within-regime AR lag).

Appendix B — Silhouette Sweep & Regime Means

K-Means silhouette by K

K	Silhouette
2	**0.4238** ✓
3	0.4048
4	0.3321
5	0.3351
6	0.3205
7	0.3022



WACMR distribution by discovered regime — 1.72 pp gap in means.

Appendix C — Pipeline Reproducibility

Stage 1
Data Ingestion
NDAP (8) + Yahoo (2)

→

Stage 2
Alignment & DB
Fri-snap | SQLite

→

Stage 3
Advanced EDA
Dist | Corr | Anomaly

→

Stage 4
Regime Discovery
PCA → K-Means K=2

→

Stage 5
Supervised ML
Walk-fwd XGBoost

→

Stage 6
NLP Corpus
75 events | Sentiment

Reproducibility checklist

- ✓ Python packages pinned to specific versions.
- ✓ Stages idempotent (run twice = same result).
- ✓ Random seeds fixed (K-Means: 42, XGBoost: deterministic).
- ✓ SQLite database committed to repo.
- ✓ Model artefacts committed (XGBoost JSON, SHAP matrix, centroids).

What this enables

- Anyone can rerun the full pipeline from raw NDAP downloads to dashboard in under an hour.
- Each stage reads only from the previous stage's persisted output — no hidden state.
- Future improvements (T-Bill yields, FinBERT) plug in at stage boundaries.